

TAMIL NADU STATE BOARD - STANDARD 12 PHYSICS VOLUME 1

CHAPTER 2: CURRENT ELECTRICITY

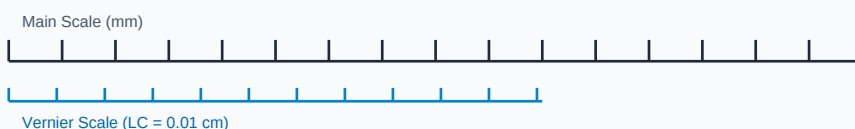
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Physics Volume 1
Chapter Name:	Current Electricity

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division

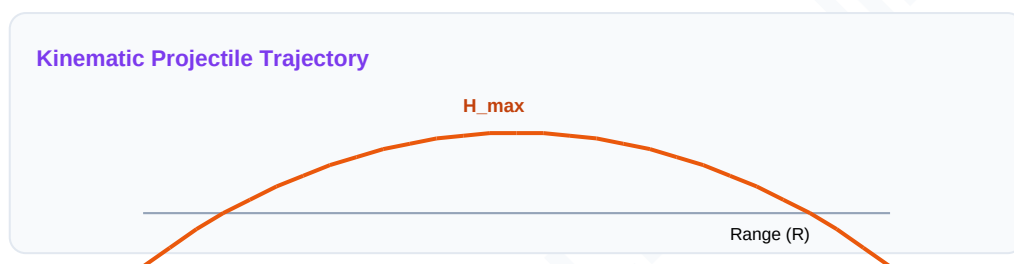


Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Drift Velocity	The average velocity acquired by free electrons in a conductor under the influence of an electric field.
Temperature Coefficient of Resistance	The fractional change in resistance of a conductor per unit change in temperature.
Superconductivity	The property of certain materials to conduct electricity with zero resistance below a transition temperature.

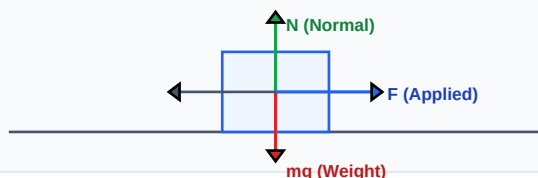
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Current: $I = n \cdot A \cdot e \cdot v_d$ | Current Density: $J = I / A = \sigma \cdot E$
- Ohm's Law: $V = I \cdot R$ | Resistance: $R = \rho \cdot l / A$
- Temperature: $R_T = R_0 \cdot (1 + \alpha \cdot \Delta T)$
- Kirchhoff's: $\sum I = 0$ (Junction) | $\sum I \cdot R = \sum E$ (Loop)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

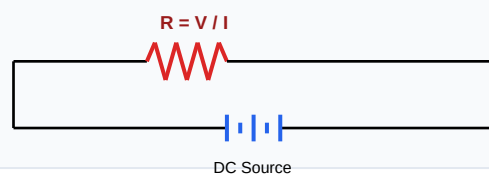
Example 1: Find resistance of a copper wire of length 10m and cross-sectional area 1 mm². (Resistivity = 1.7×10^{-8} ohm-m)

Step-by-step Solution:

Area = 10^{-6} m².

$R = \rho \cdot l / A = 1.7 \times 10^{-8} \cdot 10 / 10^{-6} = 1.7 \times 10^{-1} = 0.17$ ohms.

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Derive the balance condition of a Wheatstone Bridge using Kirchhoff's rules.

Analytical Solution Strategy:

Bridge has resistors P, Q, R, S. Let galvanometer current be I_g .

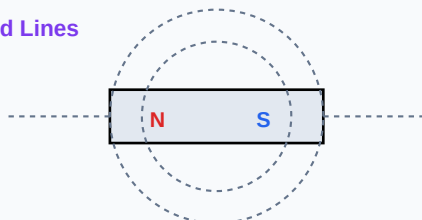
At node B: $I_1 - I_g - I_3 = 0$. At node D: $I_2 + I_g - I_4 = 0$.

Loop ABDA: $I_1 P + I_g G - I_2 R = 0$. Loop BCDB: $I_3 Q - I_4 S - I_g G = 0$.

At balance $I_g = 0 \Rightarrow I_1 = I_3$ and $I_2 = I_4$.

Equations yield $I_1 P = I_2 R$ and $I_1 Q = I_2 S$. Divide both to get $P/Q = R/S$.

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Define drift velocity and write its expression.

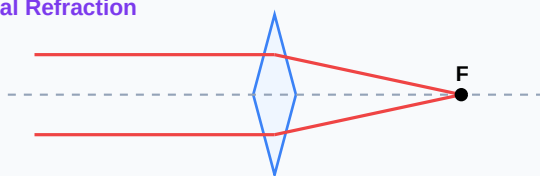
Q2 (2-Mark): State Kirchhoff's junction rule and loop rule.

Q3 (2-Mark): Explain the working of a Potentiometer for comparing EMFs.

Q4 (5-Mark): List the differences between primary and secondary electric cells.

Q5 (5-Mark): State Joule's law of heating.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Sensors

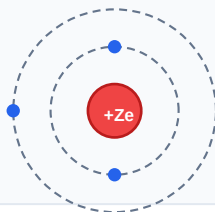
Potentiometric sensors measure micro-voltages to track cardiac activity monitors.

Application Case: Industrial safety

Fuses melt using Joule heating bounds to break grids before fire hazards occur.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

